

meme_stock_predictor/

- ├── .env # Environment variables (API keys, DB credentials)
- ├── .gitignore # Ignore .env, data/, __pycache__
- ├── README.md # Setup, usage, API docs
- ├── requirements.txt # Python dependencies
- ├── data/ # All data storage (ignored in Git)
 - ├── raw/ # Raw scraped data (CSV/JSON)
 - ├── reddit/
 - ├── twitter/
 - └── news/
 - ├── cleaned/ # Processed data (CSV/Parquet)
 - └── backups/ # Zipped monthly backups
- ├── src/
 - ├── data_collection/ # Scraping scripts
 - ├── reddit.py # PRAW/pushshift logic
 - ├── twitter.py # Tweepy/snsrape logic
 - ├── news.py # AlphaVantage/NewsAPI
 - └── utils.py # Rate-limiting, error handling
 - ├── data_processing/ # Cleaning and storage
 - ├── clean.py # Text normalization, deduplication
 - ├── storage.py # S3/PostgreSQL/InfluxDB writers
 - └── merge.py # Combine sources into master CSV
 - ├── ml/ # Model training/inference
 - ├── sentiment.py # VADER/BERT sentiment analysis
 - ├── timeseries.py # Prophet/LSTM forecasting
 - └── train.py # Hybrid model training
 - ├── api/ # FastAPI backend
 - ├── app.py # Prediction endpoints
 - ├── schemas.py # Pydantic models
 - └── alerts.py # Twilio/Discord webhooks
 - └── monitoring/ # Pipeline health
 - ├── drift.py # Evidently AI reports
 - ├── logs.py # ELK stack config
 - └── alerts.py # Sentry/PagerDuty integration

```
|—— tests/          # Unit/integration tests
|   |—— test_reddit.py
|   |—— test_sentiment.py
|   |—— conftest.py    # Fixtures
|
|—— notebooks/      # EDA and prototyping
|   |—— EDA.ipynb      # Exploratory analysis
|   |—— Model_Prototype.ipynb
|
|—— scripts/        # Utility scripts
|   |—— deploy.sh      # CI/CD deployment
|   |—— backup_data.sh # S3/GS bucket sync
|   |—— airflow_dag.py # Scheduled pipeline
|
|—— infra/          # IaC (Infrastructure as Code)
|   |—— terraform/     # AWS/GCP provisioning
|   |—— docker/        # Containerization
|       |—— Dockerfile.api
|       |—— Dockerfile.ml
|   |—— kubernetes/    # Helm charts for EKS/GKE
```